

JEE MAIN REVISION MASTER: FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 158 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Injective (One-One), Surjective (Onto) & Inverse	81 PYQs	Mappings
Very High	Domain, Range & Composite Functions	79 PYQs	Domain & Range

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Injective (One-One), Surjective (Onto) & Inverse	$\text{Injective: } f'(x) > 0 \text{ (strictly monotonic)}, \quad \text{Onto: } \text{Range}(f) = \text{Codomain}(f), \quad \text{Bijective} \iff \text{Invertible}$
Domain, Range & Composite Functions	$\text{For } \sqrt{f(x)} \implies f(x) \geq 0, \quad \text{For } \log_b(f(x)) \implies f(x) > 0, b > 0, b \neq 1, \quad x = [x] + \{x\}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Injective (One-One), Surjective (Onto) & Inverse (Very High Yield — 81 PYQs)

Core Principle: Horizontal line test, monotonicity test for injectivity, codomain matching for surjectivity, and finding function inverses.

■ **Caution / JEE Pitfall:** A function cannot be invertible unless it is BOTH one-one and onto (bijective).

■ **Speed Trick:** If $f(x)$ is continuous and strictly increasing ($f'(x) > 0$) or strictly decreasing on $\mathbb{R}$, it is always one-one.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x^2 - 3x + 23x^2 + x + 3}{x^2 + 1}$. Then f is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Let f be a polynomial function such that $\log_2(f(x)) = (\log_2(2 + \frac{2}{3}) + \frac{2}{9} + \dots \infty) \cdot \log_3(1 + \frac{f(x)}{3})$..." [Ans: 395]
- [2026 Morning Shift] Q#2 (MCQ): "The number of functions $f: 1, 2, 3, 4 \rightarrow a, b, c$, which are not onto, is :..." [Ans: B]

3.2 Domain, Range & Composite Functions (Very High Yield — 79 PYQs)

Core Principle: Determination of real domain and range, floor and fractional part functions, and domain constraints on composite functions.

■ **Caution / JEE Pitfall:** Domain of $f(g(x))$ is NOT simply the domain of g ; it requires $x \in \text{Domain}(g)$ AND $g(x) \in \text{Domain}(f)$!

■ **Speed Trick:** The range of $f(x) = a \sin x + b \cos x + c$ is $[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}]$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}((4x+2[x] / 3))$ is $[\alpha, \beta]$, then $12(\alpha+\beta)$ is equal to..." **[Ans: A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Let the domain of the function $f(x) = \log_3(\log_5(7 - \log_2(x^2 - 10x + 85))) + \sin^{-1}((3x - 7 / 17 - x))$ be $(\alpha, \beta]$. Then $\alpha + \beta$ is equal to..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}((4x+2[x] / 3))$ is $[\alpha, \beta]$, then $12(\alpha+\beta)$ is equal to :

- (A) 6 **[Correct]**
- (B) 8
- (C) 9
- (D) 4

Correct Answer: A

Solution Analysis: For the function $f(x) = \cos^{-1}((4x+2[x] / 3))$, the expression inside $\cos^{-1}$ must lie between -1 and 1. So, for domain: $-1 \leq (4x+2[x] / 3) \leq 1$ Multiplying by 3 : $-3 \leq 4x+2[x] \leq 3$ Now let $[x] = n$ where n is an integer and $n \leq x < n+1$. Then the inequality becomes $-3 \leq 4x+2n \leq 3$. Subtracting $2n$: $-3-2n \leq 4x \leq 3-2n$ Dividing by 4 : $(-3-2n / 4) \leq x \leq (3-2n / 4)$ But also, $n \leq x < n+1$. So we find possible integer values of n for which these intervals overlap. Case 1: $n = -1$ Then $[x] = -1, -1 \leq x < 0$ and $\frac{-3-2(-1)}{4} \leq x \leq \frac{3-2(-1)}{4}$

Benchmark Problem 2: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let f be a polynomial function such that $\log_2(f(x)) = (\log_2(2 + (2/3) + (2/9) + \dots \infty)) \cdot \log_3(1 + (f(x) / f(1/x)))$, $x > 0$ and $f(6) = 37$. Then $\sum_{n=1}^{10} f(n)$ is equal to ____.

Correct Answer: 395

Solution Analysis: Statement of Work : 1. $f(x)$ is a polynomial function 2. A logarithmic equation involving function f : $\log_2[f(x)] = [\log_2(2 + (2/3) + (2/9) + \dots \infty)] \cdot \log_3(1 + (f(x) / f(1/x)))$; $x > 0$ and $f(6) = 37$ 3. To evaluate the sum : $\sum_{n=1}^{10} f(n)$ $\begin{aligned} \log_2[f(x)] &= [\log_2(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty)] \cdot \log_3(1 + \frac{f(x)}{f(1/x)}) \\ &= [\log_2(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty)] \cdot \log_3(1 + \frac{f(x)}{f(1/x)}) \\ &= [\log_2(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty)] \cdot \log_3(1 + \frac{f(x)}{f(1/x)}) \end{aligned}$